

Chia-Hung Lin

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Experience

Waymo | Mountain View, CA

Nov. 2019 - Present

Hardware Test Development Engineer, LiDAR, Advanced Manufacturing

- Design, develop and build new automated test equipment for pre-production bespoke electro-mechanical LiDAR hardware for manufacturing, reliability and design validation.
- Lead production final assembly, testing, packaging and calibration for LiDAR sensor production lines with US and overseas manufacturers for Waymo's autonomous vehicles.
- Conduct failure analysis and spearhead yield improvement efforts to identify production test coverage gaps, hardware design problems and software bugs.
- Designed and implemented a Robotic Process Automation system to automate monitoring flows for Return Material Authorization (RMA), production yield and failure triage.
 - Turnaround for an improved monitoring process reduced from 24 hours to 10 minutes.
 - RMA turnaround time reduced from nondeterministic 30+ days to <10 days with >75% tasks automated.

Waymo | Mountain View, CA

July 2019 - Oct. 2019

Intern, Hardware Test Engineer, Supply Chain

- Designed robotic testing infrastructure capable of automatic parameter searching and optimization, which improved the performance of a radio wave communications link by 1000x.
- Developed a Free-Space Optical Communication system that automatically aligns to an incoming beam of laser.
- Designed an electrostatic discharge (ESD) dissipation harness and circuitry to resolve a factory issue.

Education

Johns Hopkins University | Baltimore, MD

May 2019

MSE, Robotics in Automation Science and Engineering track

National Tsing Hua University | Hsinchu, Taiwan

June 2015

BS, Program of Electrical Engineering and Computer Science (EECS)

Skills

- **Software:** Python, C++, Arduino, MATLAB, Robot Operating System (ROS)
- **Hardware:** Creo, Solidworks, Altium
- **Foreign Languages:** Fluent in Mandarin Chinese

Research & Patents

Broadband rotary joint for millimeter wave transmission | Waymo LLC

Feb 2024

- Co-inventor for **US Patent US-11916273-B1** for rotary mmWave communication link, enabling high-speed data transfer in rotating systems.
- Responsible for designing and implementing an autonomous robotic design validation and data collection system, which is used extensively to find the capable design for the mmWave link.

CoSTAR Block Stacking Dataset | Johns Hopkins University

Sept 2018 – Mar. 2019

- Accepted for **IROS 2019**: A. Hundt, V. Jain, **C. H. Lin**, C. Paxton, and G. D. Hager, "The CoSTAR Block Stacking Dataset: Learning with Workspace Constraints." Available: <https://arxiv.org/abs/1810.11714>
- Contributed to the release of the [CoSTAR Block Stacking Dataset](#) on robotic grasping with >10k attempts.
- Implemented data loader, training script, and visualization scripts to be released as a package on PyPI for Tensorflow and PyTorch. Reduced the time to load a batch of data by up to 80%.